

The mathcomp-eulerian formalization

A Stanley-style reading of permutation statistics in Rocq/MathComp

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About this document

This document is a Stanley-style reading of the [mathcomp-eulerian](#) formalization. Each definition and statement is given twice: once in ordinary mathematical English (in the spirit of Stanley *Enumerative Combinatorics I*, Chapter 1) and once as the verbatim Rocq text, with a clickable link to the line in the repository on GitHub. The intention is that a mathematician can read top to bottom like a textbook and at any point click through to verify the formal statement.

Status of this prototype. This first edition contains only Chapter 1 (Stanley §1.4: descents, Eulerian numbers, and the set-refined count $\beta_n(S)$). Chapters covering §1.3 (cycles, inversions, Foata), the q -analogues, §1.6.2 (longest alternating subsequence), and §1.6.3 (toggle action and the headline corollary) are planned but not yet written.

Conventions. The single recurring shift is 0-indexing. Stanley writes $w \in \mathfrak{S}_n$ for permutations of $\{1, \dots, n\}$ and $D(w) \subseteq \{1, \dots, n-1\}$ for the descent set; we use $\{\text{perm } 'I_n\}$ for permutations of $\{0, \dots, n-1\}$ and $\{\text{set } 'I_{n-1}\}$ for the descent set. Concretely:

$$\text{our } \{\text{perm } 'I_{n+1}\} = \text{Stanley's } \mathfrak{S}_{n+1}.$$

A complete translation table is in [docs/READING_GUIDE.md](#); a worked example through Stanley's $w = [3, 1, 4, 2]$ is in the same file.

Reading the formal text. Each Rocq block looks like this:

[\[Rocq, descent.v:25\]](#)

```
Definition descent_set s : {set 'I_n} := [set i | is_descent s i].
```

The bracketed link at the top is clickable and opens the exact line on GitHub. The body is the Rocq source verbatim; identifiers and keywords are highlighted but no mathematical substitution has been performed. If the formal block looks different from what you expected from the informal statement above it, that is exactly the kind of discrepancy worth flagging.

What is *not* included here. Full proofs are not included; only the formal *statements* are shown alongside the informal mathematics, with a one-paragraph proof sketch for each non-trivial result. The full proof is always one click away in the linked source. The `cd-index` development (`mmtree.v`, `psi_*.v`) is project-internal scaffolding not in Stanley and is not covered.

Verifying axiom-freeness. Every theorem in this document is kernel-validated by Rocq with no axioms or admits. To check this yourself for, e.g., the headline Corollary ??:

```
echo 'From mathcomp_eulerian Require Import beta_swap.
      Print Assumptions beta_alt_max.' | rocq top -R . mathcomp_eulerian
```

prints `Closed under the global context`, meaning no axiom is used anywhere in its transitive dependencies.

Chapter 1

Descents and Eulerian numbers

This chapter formalizes Stanley §1.4 (Stanley *Enumerative Combinatorics I*, 2nd ed., pp. 30–38). The central objects are the *descent set* of a permutation, the *Eulerian numbers* $A(n, k)$ that count permutations by descent count, and the set-refined count $\beta_n(S)$ that counts permutations by their full descent set.

1.1 The descent set

Definition 1.1 (Descent). Let $w = (w_1, w_2, \dots, w_n) \in \mathfrak{S}_n$. A position $i \in \{1, \dots, n-1\}$ is a *descent* of w if $w_i > w_{i+1}$.

[Rocq, descent.v:22]

```
Definition is_descent s i : bool :=
  s (widen_ord (leqnSn n) i) > s (lift_ord0 i).
```

Here s has type $\{\text{perm } 'I_{n+1}\}$ and i has type $'I_n$. The cast `widen_ord (leqnSn n) i` sends $i : 'I_n$ to the same numerical value in $'I_{n+1}$ (the position i itself); the cast `lift_ord0 i` sends i to $i + 1 : 'I_{n+1}$. Hence the formal definition reads $s(i) > s(i + 1)$, matching Definition 1.1 modulo 0-indexing.

Definition 1.2 (Descent set). The *descent set* of $w \in \mathfrak{S}_n$ is

$$D(w) = \{i \in \{1, \dots, n-1\} : w_i > w_{i+1}\}.$$

[Rocq, descent.v:25]

```
Definition descent_set s : {set 'I_n} := [set i | is_descent s i].
```

$D(w)$ is a subset of $\{1, \dots, n-1\}$ in Stanley; in our types it is a subset of $\{0, \dots, n-1\} = 'I_n$. The correspondence is Stanley's $i \leftrightarrow$ our $(i - 1)$.

Example 1.3. Take $w = [3, 1, 4, 2] \in \mathfrak{S}_4$. Then $D(w) = \{1, 3\}$ since $w_1 = 3 > 1 = w_2$ and $w_3 = 4 > 2 = w_4$. In our types, the corresponding $s : \{\text{perm } 'I_4\}$ has $s\ 0 = 2$, $s\ 1 = 0$, $s\ 2 = 3$, $s\ 3 = 1$ (values shifted to $\{0, 1, 2, 3\}$), and `descent_set s = {0, 2} : {set 'I_3}`.

Definition 1.4 (Descent number, ascent number). $\text{des}(w) = |D(w)|$ and $\text{asc}(w) = (n - 1) - \text{des}(w)$.

[Rocq, descent.v:27]

```
Definition des s : nat := #|descent_set s|.
```

[Rocq, descent.v:29]

Definition `asc s : nat := n - des s.`

In the formal definition, n is the number of descent positions (Stanley's $n - 1$). So our `asc s = n - des s` matches Stanley's $(n - 1) - \text{des}(w)$.

Lemma 1.5 (Descent count is bounded). $\text{des}(w) \leq n - 1$ for every $w \in \mathfrak{S}_n$.

[Rocq, descent.v:38]

Lemma `des_le s : des s ≤ n.`

Proof sketch. $D(w) \subseteq \{0, \dots, n - 1\}$ which has n elements; the cardinality is at most n . $\square$

Lemma 1.6 (Identity has no descents). $\text{des}(\text{id}) = 0$.

[Rocq, descent.v:44]

Lemma `des_id : des (1 : {perm 'I_n.+1}) = 0.`

Proof sketch. For the identity, $s i = i$ at every position, so $s(i) = i < i + 1 = s(i + 1)$ — no descent slot. $\square$

1.2 Reversal symmetry

A useful symmetry: reversing a permutation flips ascents and descents at corresponding positions.

Definition 1.7 (Reverse permutation). Let $w \in \mathfrak{S}_n$ and define w^{rev} by $w_i^{\text{rev}} = w_{n+1-i}$.

[Rocq, descent.v:80]

Definition `rev_perm_ord : {perm 'I_n.+1} := perm (@rev_ord_inj n.+1).`

[Rocq, descent.v:85]

Definition `rev_perm s : {perm 'I_n.+1} := rev_perm_ord * s.`

`rev_perm_ord` is the position-reversal involution $i \mapsto n - i$; `rev_perm s` pre-composes s with it, giving Stanley's w^{rev} .

Lemma 1.8 (Descents of the reverse). *Position i is a descent of w^{rev} iff position $n - i$ is not a descent of w :*

$$i \in D(w^{\text{rev}}) \iff (n - i) \notin D(w).$$

[Rocq, descent.v:90]

Lemma `is_descent_rev s (i : 'I_n) : is_descent (rev_perm s) i = ~ is_descent s (rev_ord i).`

Proof sketch. Direct calculation: $(\text{rev_perms}) j = s(n - j)$, and the comparison $s(n - i) > s(n - i - 1)$ is the negation of $s(n - i - 1) > s(n - i)$. $\square$

1.3 Eulerian numbers

Definition 1.9 (Eulerian number). The *Eulerian number* $A(n, k)$ is the number of permutations of $\{1, \dots, n\}$ with exactly k descents:

$$A(n, k) = \#\{w \in \mathfrak{S}_n : \text{des}(w) = k\}.$$

[Rocq, eulerian.v:14]

```
Definition eulerian (n k : nat) : nat :=
  #| [set s : {perm 'I_n.+1} | des s == k] |.
```

Off-by-one alert. The Rocq parameter n is "permutation size minus one." Concretely, `eulerian n k` counts perms of $\{\text{perm } 'I_{n+1}\}$, which are Stanley's $\mathfrak{S}_{n+1}$. Hence

$$\text{eulerian } n \ k = A(n+1, k).$$

For instance, `eulerian 3 1` corresponds to Stanley's $A(4, 1) = 11$ (Stanley table p. 36).

Lemma 1.10 (Row sum). $\sum_{k=0}^{n-1} A(n, k) = n!$

[Rocq, eulerian.v:31]

```
Lemma eulerian_row_sum_fact n :
  \sum_ (k < n.+1) eulerian n k = n.+1'!.
```

Proof sketch. The summands partition $\mathfrak{S}_{n+1}$ by descent count; the total is $|\mathfrak{S}_{n+1}| = (n+1)!$. $\square$

Lemma 1.11 (Out-of-range vanishing). $A(n, k) = 0$ for $k \geq n$.

[Rocq, eulerian.v:34]

```
Lemma eulerian_out_of_range n k :
  n < k → eulerian n k = 0.
```

Proof sketch. A permutation has at most n descent positions (Lemma 1.5); none can have $k > n$ descents. $\square$

Lemma 1.12 (The unique perm with no descents is the identity). *If $\text{des}(w) = 0$ then $w = \text{id}$.*

[Rocq, eulerian.v:41]

```
Lemma des0_id n (s : {perm 'I_n.+1}) : des s = 0 → s = 1%g.
```

Proof sketch. $\text{des}(s) = 0$ means $s(i) < s(i+1)$ for every adjacent pair i , so s is strictly increasing. The only strictly increasing function $'I_{n+1} \rightarrow 'I_{n+1}$ is the identity. $\square$

1.4 The insert-max bijection

A central tool for the Eulerian recurrence is the bijection that inserts the maximum value $n+1$ at a chosen position into a permutation of $\{1, \dots, n\}$.

Definition 1.13 (Insert-max). For $t \in \mathfrak{S}_n$ and $p \in \{1, \dots, n+1\}$, let $\text{insert_max}(t, p) \in \mathfrak{S}_{n+1}$ be the permutation obtained by inserting the value $n+1$ at position p in the one-line form of t .

[Rocq, eulerian.v:134-157]

```
Definition insert_max_fun (i : 'I_n.+2) : 'I_n.+2 :=
  match unlift p i with
  | Some j ⇒ widen_ord (leqnSn _) (t j)
  | None ⇒ ord_max
  end.

Definition insert_max_perm : {perm 'I_n.+2} := perm
  insert_max_fun_inj.
```

At position p , the output is `ord_max` = $n+1$. At any other position `lift p j` (i.e., positions $\neq p$, parameterized by $j : 'I_{n+1}$), the output is $t j$ cast back to $'I_{n+2}$.

Definition 1.14 (Extract-max). The inverse: given $\sigma \in \mathfrak{S}_{n+1}$ with $\sigma(p) = n + 1$ for some position p , remove that value and standardise to obtain $t \in \mathfrak{S}_n$.

[Rocq, eulerian.v:185-203]

```
Definition extract_max_fun (j : 'I_n.+1) : 'I_n.+1 :=
  odflt j (unlift ord_max (s (lift p j))).
```

```
Definition extract_max_perm : {perm 'I_n.+1} := perm
  extract_max_fun_inj.
```

Theorem 1.15 (Insert-max is a bijection). *The map $(t, p) \mapsto \text{insert_max}(t, p)$ is a bijection $\mathfrak{S}_n \times \{1, \dots, n + 1\} \rightarrow \mathfrak{S}_{n+1}$.*

[Rocq, eulerian.v:437]

```
Lemma insert_max_perm_bij n :
  ∀ sigma : {perm 'I_n.+2},
  ∃! tp : {perm 'I_n.+1} * 'I_n.+2,
    sigma = insert_max_perm tp.1 tp.2.
```

Proof sketch. Given σ , the unique p is $\sigma^{-1}(n+1)$ and $t = \text{extract_max}(\sigma, p)$. The two roundtrips are direct calculations. $\square$

1.5 Eulerian recurrence

Theorem 1.16 (Eulerian recurrence). *For $n \geq 1$ and $0 \leq k \leq n$,*

$$A(n + 1, k) = (k + 1) A(n, k) + (n - k + 1) A(n, k - 1).$$

[Rocq, eulerian.v:458]

```
Theorem eulerian_rec n k :
  eulerian n.+1 k =
    (k.+1) * eulerian n k + (n.+1 - k) * eulerian n k.-1.
```

Proof sketch. Using the insert-max bijection, classify each $\sigma \in \mathfrak{S}_{n+1}$ by the position p of $n + 1$. There are $k + 1$ ascent positions and $n - k$ descent positions where inserting $n + 1$ preserves the descent count; inserting at the other $n - k + 1$ slots increases it by one. Summing recovers the recurrence. $\square$

1.6 The set-refined count β

The descent count is a coarse invariant. The full descent *set* carries more information; the count of permutations with a fixed descent set is a finer object.

Definition 1.17 (β count). For a subset $S \subseteq \{1, \dots, n - 1\}$, the count $\beta_n(S)$ is the number of permutations of $\{1, \dots, n\}$ whose descent set is exactly S :

$$\beta_n(S) = \#\{w \in \mathfrak{S}_n : D(w) = S\}.$$

[Rocq, beta.v:19]

```
Definition beta (n : nat) (D : {set 'I_n}) : nat :=
  #|[{set sigma : {perm 'I_n.+1} | descent_set sigma == D}]|.
```

Off-by-one alert. As with `eulerian`, the Rocq parameter n is "perm size minus one." So `beta n D` for $D : \{\text{set } 'I_n\}$ equals Stanley's $\beta_{n+1}(D)$, with D interpreted as a subset of $\{0, \dots, n-1\}$ that corresponds to Stanley's 1-indexed $S \subseteq \{1, \dots, n\}$.

Lemma 1.18 (β partitions Eulerian). *Summing $\beta_n(S)$ over all S of cardinality k recovers the Eulerian number:*

$$\sum_{|S|=k} \beta_n(S) = A(n, k).$$

[Rocq, beta.v:124]

```
Lemma beta_eulerian n k :
  \sum_ (D : {set 'I_n} | #|D| == k) beta D = eulerian n k.
```

Proof sketch. The sets $\{\sigma : D(\sigma) = D\}$ for D ranging over subsets of cardinality k partition the set of permutations with k descents. The cardinalities sum. $\square$

Lemma 1.19 (β is reversal-invariant). *For $S \subseteq \{1, \dots, n-1\}$, let $S^{\text{rev}} = \{n-i : i \in S\}$. Then $\beta_n(S^{\text{rev}}) = \beta_n(S^c)$, where S^c is the complement in $\{1, \dots, n-1\}$.*

[Rocq, beta.v:140]

```
Lemma beta_rev n (D : {set 'I_n}) :
  beta [set rev_ord i | i in D] = beta (~: D).
```

Proof sketch. Reversal $w \mapsto w^{\text{rev}}$ is a bijection on $\mathfrak{S}_n$; by Lemma 1.8 it maps descent set D to $\{n-i : i \notin D\}$, which is the complement of $\{n-i : i \in D\}$. Counting on both sides matches. $\square$

1.7 What this chapter doesn't cover

This chapter establishes the basic objects (D , des , A , β) and the foundational identities. The headline of the formalization,

$$\beta_n(S) \leq E_n \quad \text{with equality iff } S \text{ is alternating} \quad (\text{Stanley Cor 1.6.5})$$

requires the toggle-action machinery of §1.6.3 and is the subject of a future chapter.

Appendix A

Source map

A complete file index is in `docs/READING_GUIDE.md` in the [mathcomp-eulerian](#) repository. Files covered (or planned) by this document, in Stanley reading order:

descent.v, **eulerian.v**, **beta.v** Stanley §1.4 (this chapter).

inversions.v Stanley §1.3.3 (planned).

cycles.v Stanley §1.3.1–2 (planned).

foata.v Stanley §1.3.4 (planned).

qfact.v, **qeul.v** Stanley §1.4 q -analogues (planned).

altsub.v Stanley §1.6.2 (planned).

beta_omega.v, **beta_swap.v** Stanley §1.6.3 (planned).

experimental/reflection.v Stanley §1.6.4 (planned, contains 1 named admit).